

FlowPortal: Residual-Corrected Flow for Training-Free Video Relighting and Background Replacement

Anonymous CVPR submission

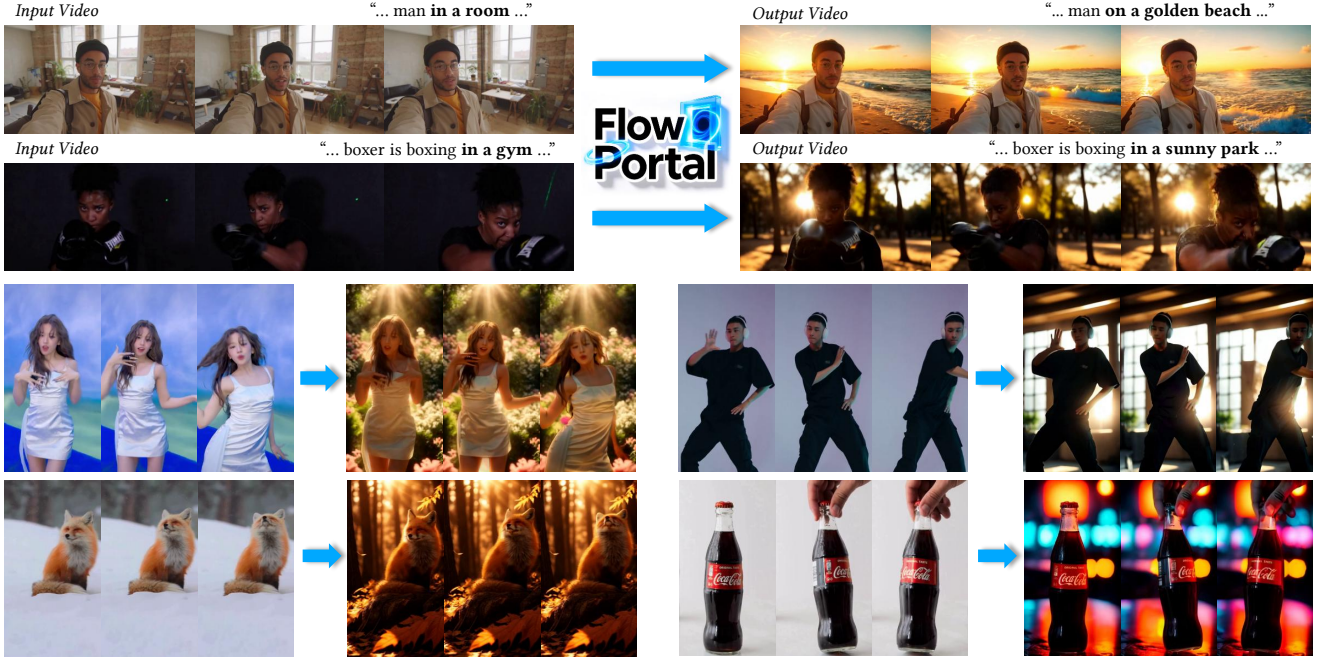


Figure 1. We propose a novel training-free **FlowPortal** framework for efficient video background replacement and foreground relighting.

Abstract

Video relighting with background replacement is a challenging task critical for applications in film production and creative media. Existing methods struggle to balance temporal consistency, spatial fidelity, and illumination naturalness. To address these issues, we introduce FlowPortal, a novel training-free flow-based video relighting framework. Our core innovation is a Residual-Corrected Flow mechanism that transforms a standard flow-based model into an editing model, guaranteeing perfect reconstruction when input conditions are identical and enabling faithful relighting when they differ, resulting in high structural consistency. This is further enhanced by a Decoupled Condition Design for precise lighting control and a High-Frequency Transfer mechanism for detail preservation. Additionally, a masking strategy isolates foreground relighting from background pure generation process. Experiments demonstrate that FlowPortal achieves superior performance in temporal coherence, structural preservation, and lighting realism, while maintaining high efficiency.

1. Introduction

Harmonious lighting between the foreground and the background plays a crucial role in determining the realism, visual quality, and aesthetic appeal of a video. The capability to generate a new background that harmonizes with the foreground using adapted lighting, known as the video relighting task with background replacement, has attracted increasing attention due to its broad range of applications in film production, virtual photography, commercial display, and creative media editing. This technology effectively functions as a visual portal, dramatically reducing the need for on-site shooting by enabling flexible scene composition and stylized video generation. Furthermore, this capability holds significant potential for future world-to-world portal applications that seamlessly bridge different visual worlds.

Although promising, video relighting with background replacement poses significant challenges due to the need to maintain video quality, temporal consistency, and subject fidelity under new lighting conditions. Pursuing one goal often compromises others, leading to temporal flick-

040	ering, loss of detail, unrealistic lighting, or prohibitive	093
041	computational cost. For example, training-based meth-	094
042	ods [4, 7, 12, 15, 17, 28] are resource-intensive and struggle	095
043	with creating paired dataset and balancing between illumi-	096
044	nation richness and content fidelity. Meanwhile, training-	097
045	free methods like AnyPortal [5] and Light-A-Video [34]	098
046	that combine pretrained image relighting and video gener-	099
047	ation models suffer from temporal inconsistency caused by	100
048	per-frame relighting as well as misalignment between input	101
049	and output video due to weak condition control and limited	102
050	capability for video models.	103
051	We argue that these issues arise from the absence of a co-	104
052	hesive framework that systematically disentangles and con-	
053	trols the core elements of a video: structure, motion, and	
054	illumination. To bridge this gap, we introduce a new Flow-	
055	Portal framework built upon a condition-aware editing	
056	model through two dedicated mechanisms. First, we pro-	
057	pose a Residual-Corrected Flow with High-Frequency	
058	Transfer for structure and detail preservation. Second, De-	
059	coupled Condition Design is introduced for precise light-	
060	ing control. Together, these advancements provide a uni-	
061	fied solution that addresses the long-standing challenges of	
062	video relighting: temporal continuity, spatial fidelity, illu-	
063	mination naturalness, and operational efficiency.	
064	Specifically, our key insight is to exploit the principle	
065	of Condition Consistency that every change in the output	
066	is driven by a change in the target condition (<i>i.e.</i> , illumina-	
067	tion). It implies a perfect reconstruction when there is no	
068	change in condition. Based on this principle, we introduce	
069	a Residual-Corrected Flow mechanism that rephrases a typ-	
070	ical flow-based generation model to a novel training-free	
071	editing model. It ensures structural consistency by directing	
072	the flow towards perfect reconstruction when the target and	
073	source conditions are identical, so that subsequent changes	
074	in the conditions like illumination variation can be reflected	
075	in the video output. Our Residual-Corrected Flow allows	
076	for reusing velocity prediction across timesteps to enhance	
077	efficiency, and it processes videos holistically, avoiding per-	
078	frame processing to ensure temporal continuity.	
079	To further strengthen the model’s fidelity and controlla-	
080	bility, we introduce three key designs: First, we propose a	
081	Decoupled Condition Design with illumination-specific and	
082	agnostic visual-textual condition inputs. This decomposi-	
083	tion provides a stable, sufficient, and directional guidance	
084	that engineeringly enforces Condition Consistency. Sec-	
085	ond, we introduce High-Frequency Transfer, which adheres	
086	to Condition Consistency by injecting illumination-agnostic	
087	details to solidify fidelity. Third, we employ masks to iso-	
088	late and precisely relight the foreground while maintaining	
089	a pure background generation process, allowing for a high-	
090	quality, contextually natural background replacement. In	
091	summary, our contributions are threefold:	
092	• We propose a novel training-free FlowPortal framework	
	for coherent and efficient video relighting and back-	093
	ground replacement. By introducing Residual-Corrected	094
	Flow and Decoupled Condition Design, our framework	095
	ensures structural consistency between source and target	096
	videos, as well as illumination naturalness.	097
	• We introduce a High-Frequency Transfer mechanism	098
	within the Residual-Corrected Flow, improving fine de-	099
	tail consistency under new lighting conditions.	100
	• We propose Masked Residual-Corrected Flow and	101
	Masked High-Frequency Transfer to isolate foreground	102
	and background processing, enabling high-quality, con-	103
	textually natural background replacement.	104
	2. Related Works	105
	Image Relighting and Background Replacement. Recent	106
	advances in generation models have led to successful im-	107
	age relighting methods [9, 19, 21, 26, 31]. Total Relight-	108
	ing [19] and SwitchLight [9] train neural networks to pre-	109
	dict surface properties, namely normals and albedo, which	110
	are then used to recompute illumination. Relightful Har-	111
	monization [21] fine-tunes an image diffusion model con-	112
	ditioned on the background to achieve foreground relight-	113
	ing. As the current state-of-the-art method in image relight-	114
	ing, IC-Light [31] simultaneously performs background re-	115
	placement and illumination harmonization by concatenat-	116
	ing the input noise with a foreground condition before feed-	117
	ing it into a diffusion model trained with a light transport	118
	consistency objective to ensure a coherent composite.	119
	Relighting in Video. With the maturation of image relight-	120
	ing techniques, researchers have begun to focus on the task	121
	of video relighting [4–6, 29, 34]. Training-based methods	122
	such as IllumiCraft [12], UniRelight [7], RelightVid [4],	123
	Lumen [28], Lux Post Facto [17], and TC-Light [15] per-	124
	form video relighting by constructing datasets and train-	125
	ing video-conditional diffusion models. However, collect-	126
	ing large-scale high-quality paired video relighting datasets	127
	is difficult and requires high costs for training. Moreover,	128
	video models struggle to balance between the lighting con-	129
	ditions and the video foreground conditions. As a result,	130
	the trained models tend to produce lighting results with in-	131
	sufficient richness and diversity in illumination, and lacking	132
	fidelity under complex conditions.	133
	Some works pursue training-free approaches for effi-	134
	ciency. AnyPortal [5] and Light-A-Video [34] both adopt	135
	training-free (zero-shot) strategies, combining the capabili-	136
	ties of IC-Light and video diffusion generation models dur-	137
	ing inference. AnyPortal introduces cross-frame attention	138
	in IC-Light and applies a Refinement Projection Algorithm	139
	in the video model to ensure consistency, while Light-A-	140
	Video uses Consistent Light Attention in IC-Light and em-	141
	ploys a Progressive Light Fusion strategy in the video dif-	142
	fusion process to maintain lighting generation and consis-	143
	tency. However, in these two training-free methods, IC-	144

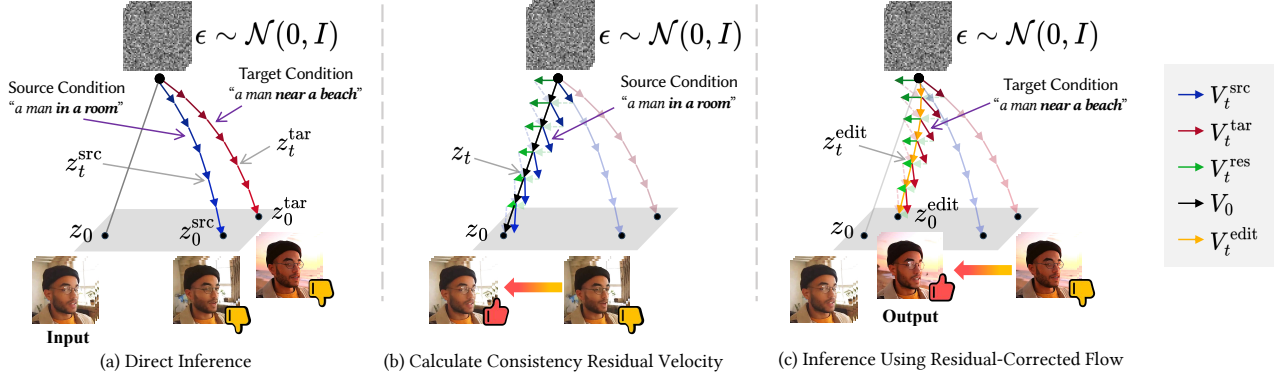


Figure 2. **Illustration of the proposed Residual-Corrected Flow.** (a) The **Naive Edit Flow** builds denoising trajectories under source and target conditions using the same noise ϵ . When applied to a real input video z_0 , the mismatch between z_0^{src} and z_0 violates Stability under Identity. (b) **Consistency Residual Velocity** V_t^{res} is constructed as the difference between the ideal restoration path V_0 and the predicted source flow V_t^{src} , aligning the generated z_0^{src} with z_0 . (c) **Residual-Corrected Flow** combines V_t^{tar} and V_t^{res} to perform reliable video relighting that preserves identity consistency (e.g., mouth shape, glasses reflection) while enabling directional condition change. The purple arrows indicate the condition to guide the velocity calculation. For simplicity, we omit the reference frame and structural conditions.

Light is still used to relight per frame individually, introducing frame discontinuities that cannot be fully corrected by the video model. In addition, due to the lack of sufficient control in the video model, the relighting results may exhibit inconsistencies in structure and motion compared to the original video. Overcomplicated pipelines also make these methods less inference-efficient.

Our method is also training-free. Rather than per-frame IC-Light processing, we adopt holistic relighting to achieve better temporal consistency. Moreover, by introducing the proposed Residual-Corrected Flow and High-Frequency Transfer, we better address structural and detail consistency. To achieve stronger conditional control, we propose a Decoupled Condition Design to enhance the model’s ability to respond to given conditions. We further achieve higher efficiency by reusing residual predictions for acceleration.

3. Method

3.1. Preliminary: Flow Models

In flow-based generative models [10, 13, 14, 24], data generation process is modeled as a learned continuous flow that gradually transforms a random noise into a data sample¹.

Formally, let z_t denote the latent variable at timestep (or noise level) $t \in [0, 1]$, where $t = 1$ corresponds to the starting point of the generation process with $z_1 \sim \mathcal{N}(0, I)$, and $t = 0$ indicates the final clean data sample z_0 . Let c represent the conditional information. The model F_θ predicts a velocity (or flow) vector field V_t^c at each noise level t :

$$V_t^c(z_t) = F_\theta(z_t, t, c). \quad (1)$$

¹For simplicity, we omit the details regarding the latent space. The video is mapped into the latent space through a VAE.

V_t^c therefore defines how the latent variable evolves continuously from noise to data over time. This generation process follows an ordinary differential equation (ODE) form:

$$\frac{dz_t}{dt} = V_t^c(z_t). \quad (2)$$

In practice, this continuous process is discretized into $N + 1$ steps $\{t_0, t_1, \dots, t_N\}$, with $0 = t_0 < t_1 < \dots < t_N = 1$. Starting from an initial latent $z_{t_N} = z_1 \sim \mathcal{N}(0, I)$, the model iteratively applies the discrete update rule

$$z_{t_{i-1}} = z_{t_i} + (t_i - t_{i-1}) V_{t_i}^c(z_{t_i}), \quad (3)$$

for N steps from t_N to t_0 , progressively refining the latent until it reaches $z_{t_0} = z_0$ that lies in the data manifold.

3.2. Condition Consistency

Condition Consistency describes a fundamental property of an ideal editing model: changes in the output are driven by changes in the input conditions. It suggests two core principles: **1)** When the editing condition changes, a corresponding and perceivable change should manifest in the output; **2)** When the editing condition remains identical, the output should be a faithful replica of the input. This principle is critically important in the specific context of video relighting. Specifically, it requires that

- **Directional Change:** The generated video should differ from the input only in lighting if the conditions differ only in lighting. All other aspects, including object structure and motion dynamics, must be perfectly preserved.
- **Stability under Identity:** In the extreme case where the target and source conditions are identical, an ideal model must reproduce the source video exactly.

It guarantees a faithful mapping of the condition signal, which neither omits required changes nor introduces extraneous ones, thereby ensuring predictable outcomes and high

visual fidelity. To this end, we aim to design a reliable video relighting model enforced with Condition Consistency.

3.3. Residual-Corrected Flow with Decoupled Condition for Reliable Video Relighting

Naive Edit Flow. We begin with a simple edit flow. We first sample a fixed Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and build two denoising trajectories over ϵ under the source condition “src” (e.g., a prompt describing the original video) and the target condition “tar” (e.g., a prompt describing the target edited video) following Eq. (1)-(3). This results in two distinct flows denoted as V_t^{src} and V_t^{tar} , respectively. The noise variable evolves along the flows from $t = 1$ to 0, producing two outputs: z_0^{src} and z_0^{tar} , as shown in Fig. 2(a). Since two flows share the same ϵ , if “tar”=“src”, then $z_0^{\text{tar}} = z_0^{\text{src}}$. Clearly, the editing from a synthetic video z_0^{src} to z_0^{tar} satisfies the property of “Stability under Identity”. However, issues arise when applying this edit flow to a real input video z_0 . Due to inherent randomness, model capability limitations, and insufficient information in condition “src”, the generated video z_0^{src} often does not exactly match z_0 . This violates Stability under Identity: when “tar”=“src”, the output $z_0^{\text{tar}} \neq z_0$.

Residual-Corrected Flow. Based on the above analysis, our key idea is to simultaneously adjust both z_0^{src} and z_0^{tar} , pulling z_0^{src} to be identical to z_0 , so that the adjusted z_0^{tar} , which we denote as z_0^{edit} , can reflect directional change to z_0 . First, we construct a restoration path from the noise ϵ to the source video z_0 . This path corresponds to a velocity defined as:

$$V_0 = \frac{z_0 - \epsilon}{1 - 0}. \quad (4)$$

If the denoising process strictly follows this velocity, the noise ϵ will be exactly transformed back into z_0 , and the intermediate result z_t at timestep t satisfies

$$z_t = (1 - t)z_0 + t\epsilon. \quad (5)$$

However, in practice, the model predicts a velocity $V_t^{\text{src}}(z_t) \neq V_0$ at z_t . To align this predicted flow with the ideal restoration path, we build a **Consistency Residual Velocity** V_t^{res} at each timestep, defined as:

$$V_t^{\text{res}}(z_t) = V_0 - V_t^{\text{src}}(z_t). \quad (6)$$

After obtaining V_t^{res} shown in Fig. 2(b), we can perform denoising starting from ϵ using the combined flow $V_t^{\text{src}}(z_t) + V_t^{\text{res}}(z_t) = V_0$. Following this adjusted flow trajectory ensures an accurate reconstruction of z_0 from ϵ .

To achieve video transfer under the target condition, we introduce a **Residual-Corrected Flow** based on the Consistency Residual Velocity. Specifically, given the model-predicted flow V_t^{tar} under the target condition and V_t^{res} , we define the final Residual-Corrected Flow as:

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + V_t^{\text{res}}(z_t). \quad (7)$$

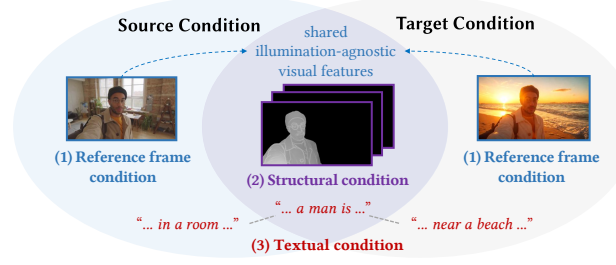


Figure 3. **Decoupled Condition Design.** The source and target conditions share identical illumination-agnostic information, differing only in their illumination-specific information.

Starting from the same initial Gaussian noise $z_1^{\text{edit}} = \epsilon$, we then perform the denoising process following V_t^{edit} from $t = 1$ to 0, shown in Fig. 2(c). The final result of this process is our target relit video z_0^{edit} .

It is not hard to prove that when “tar”=“src”, then $V_t^{\text{edit}} = V_0$ and $z_0^{\text{tar}} = z_0$, thereby satisfying Stability under Identity. Our next step is to enforce “Directional Change” property to achieve an overall Condition Consistency.

Decoupled Condition Design. To strengthen Directional Change, we introduce a Decoupled Condition Design. Our condition input is carefully crafted using both illumination-specific and agnostic features to enable the model to more effectively identify what and where to make edits. As shown in Fig. 3, it comprises:

- **Reference frame condition.** We use an I2V model as F_θ to accept a reference frame. For the source condition, the reference frame is the first frame of the input video. For the target condition, it is the first frame edited by the powerful image relighting model, IC-Light [31], to provide a robust visual anchor that guarantees both illumination naturalness and spatial fidelity.
- **Structural condition.** We employ a pretrained ControlNet [1, 30] on a weighted fusion of the depth and edge maps extracted from the input video as the illumination-agnostic structural condition. It is shared by both source and target conditions, ensuring the preservation of structural content during editing.
- **Textual condition.** We use illumination-specific textual prompts, where the source and target prompts differ only in their descriptions of background and lighting, while sharing the same foreground content.

This decomposition provides a stable, sufficient, and disentangled guidance signal to enforce Condition Consistency.

3.4. High-Frequency Transfer

To further enhance the detail consistency between the generated target video and the original source video, we aim to inject a portion of high-frequency information from the source video into the generation process of the target video.

Specifically, let $\text{HF}(X)$ and $\text{LF}(X)$ denote the high-frequency and low-frequency components of the video

X obtained by Fourier decomposition satisfying $X = \text{HF}(X) + \text{LF}(X)$. Then, during each generation step of the target video z_t^{edit} , we perform the following replacement:

$$z_t^{\text{edit}} \leftarrow \text{LF}(z_t^{\text{edit}}) + \lambda \cdot \text{HF}(z_t) + (1 - \lambda) \cdot \text{HF}(z_t^{\text{edit}}) \quad (8)$$

where z_t is the source video at step t and λ controls the proportion of high-frequency information injected. A larger λ strengthens the preservation of fine details from the source video but may limit the model’s ability to fully adapt to the target illumination condition; conversely, a smaller λ allows more flexibility in relighting but may reduce structural and textural consistency.

Note that the injection maintains Stability under Identity since when reconstructing z_0 , transferring high-frequency from z_t to z_t itself does not alter itself.

3.5. High-Quality Background Generation with Masking Mechanism

Residual-Corrected Flow and High-Frequency Transfer may introduce unwanted background details from source video into the result. To mitigate this conflict during new background generation, we use the mask to disentangle the foreground and background regions. Let M denote the mask of the foreground region extracted from the original video. Then, the Masked Residual-Corrected Flow is

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + M \cdot V_t^{\text{res}}(z_t). \quad (9)$$

Similarly, we apply the Masked High-Frequency Transfer only to the foreground region:

$$z_t^{\text{edit}} \leftarrow \text{LF}(z_t^{\text{edit}}) + \lambda M \cdot \text{HF}(z_t) + (1 - \lambda M) \cdot \text{HF}(z_t^{\text{edit}}). \quad (10)$$

The structural condition of our Decoupled Condition Design is modified accordingly. We utilize a combination of the previously extracted features (*i.e.*, depth and edge), but only for the foreground region, leaving the background unconstrained to allow new content generation.

Using this mask mechanism, we can easily preserve the foreground details while generating the background unperturbedly without any interference from the source video. Note that this mask mechanism permits certain regions to violate the Stability under Identity property, allowing for more flexible editing.

3.6. Comparison to FlowEdit in Video Relighting

Inversion and FlowEdit also aim for editing based on reconstruction. Although inversion [8, 18, 22, 25] is commonly used in image generation models for editing, it remains both time-consuming and imprecise in video models. The imprecision also violates the Stability under Identity, resulting in poor Condition Consistency. By comparison, our method is inversion-free, which is partially inspired by the recent approach, FlowEdit [11]. It offers a novel path for image

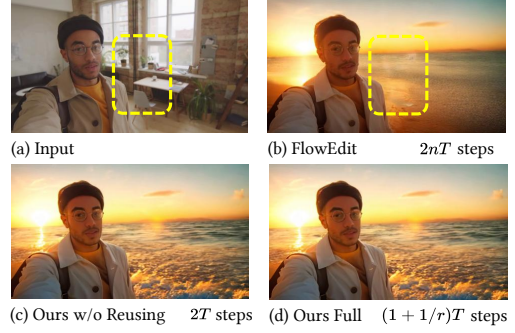


Figure 4. **Comparison with FlowEdit.** (a) Input video. (b) FlowEdit produces blurry outputs, and suffers from ghosting artifacts due to the interference from the original background (yellow region). (c)(d) Our residual reusing strategy effectively reduces the number of prediction steps with negligible quality degradation.

editing that satisfies Stability under Identity without inversion. However, it has obvious limitations when applied to the video relighting task.

In FlowEdit, starting with the source image $z_1^{\text{edit}} = z_0$, the target image z_t^{edit} evolves along the flow V_t^{edit} :

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) - V_t^{\text{src}}(z_t), \quad (11)$$

$$z_t = (1 - t)z_0 + t\epsilon_t, \quad (12)$$

$$z_t^{\text{pred}} = z_t + z_t^{\text{edit}} - z_0, \quad (13)$$

from $t = 1$ to 0, where $\epsilon_t \sim \mathcal{N}(0, I)$. Our design differs from FlowEdit in two aspects: 1) FlowEdit samples n different Gaussian noises at each timestep while we propose to use a single Gaussian noise ϵ throughout the whole editing process. 2) FlowEdit constructs an edit flow from the source video $z_1^{\text{edit}} = z_0$ to target result z_0^{edit} . We rephrase this process and build a new generation flow from noise $z_1^{\text{edit}} = \epsilon$ to the target z_0^{edit} by tracking z_t^{pred} in FlowEdit, just as the standard diffusion denoising process.

It can be proved that if FlowEdit samples a fixed noise across steps, it is theoretically equivalent to our method without masking mechanism. However, it is the above two straightforward changes that make a big difference, establishing key advantages of our approach over FlowEdit:

- **Clear Generation Ability.** To ensure stable editing, FlowEdit independently samples multiple Gaussian noises and averages V_t^{edit} at each timestep, which inevitably causes blur in the generated background as in Fig. 4 (b). Our fixed noise prevents blur, significantly improving the generation quality.
- **Acceleration.** Our method and FlowEdit require computing two velocities, doubling the denoising steps from T to $2T$ ($2nT$ for FlowEdit if averaging $n \geq 1$ velocities per step). The benefit we offer is that the proposed consistency residual velocity relies solely on the fixed noise and source condition, thus it is stable across steps and can be reused. Our experiments have shown that applying the

same residual velocity every r steps has little impact on quality, while it decreases the total steps to $(1 + 1/r)T$, as shown in Fig. 4 (c)(d). In contrast, FlowEdit requires new random noises and target-condition denoising for each edit flow calculation, which prevents reusability.

- **Spatial Controllability.** The edit flow of FlowEdit starts from the source video, which cannot handle pure generation within specific regions. The background from the source video will interfere with the new background generation as in Fig. 4 (b). By comparison, our method, starting from pure noises, enables foreground masking for editing while leaving the background for pure generation.
- **Decoupled Condition.** Additionally, we propose a Decoupled Condition Design to combine the reference frame and structural conditions, which helps the model achieve better Condition Consistency.

4. Experiments

4.1. Experimental Setup

Implementation details. We implement our method based on Wan2.1 [1, 24], a widely used state-of-the-art open-sourced video diffusion model. For the foreground mask M , we use BiRefNet [33] to generate the mask for the first frame and MatAnyone [27] to propagate it throughout the entire video. We downsample M to match the video shape in the latent space for masking operation. For the Decoupled Condition Design, the reference frame is generated by IC-Light [31], and the structural information is obtained by combining the HED, depth, and Canny maps [30], followed by masking with M . For Residual-Corrected Flow, we adopt $T = 50$ timesteps for the generation process and reuse the Consistency Residual Velocity every $r = 10$ steps, resulting in a total of 55 steps. For High-Frequency Transfer, we set the frequency decomposition threshold to 0.8 and the injection intensity to $\lambda = 0.5$. The experiments are conducted on a single 80 GB NVIDIA A100 GPU. Representative results are illustrated in Fig. 1 and Fig. 7.

Baseline methods. We compare our method with recent video relighting approaches. Training-free methods such as AnyPortal [5] and Light-A-Video [34] apply the image relighting model IC-Light frame by frame and integrate the results into video generation models. Training-based methods like Lumen [28] and TC-Light [15] are trained on paired video datasets constructed for the relighting task. The basic information for these methods is summarized in Tab. 1. For fairness, we compare only with methods that support new background generation, excluding RelightVid [4] and Lux Post Facto [17] since they are not open-sourced.

Metrics. To comprehensively evaluate the quality of relighted videos, we assess our results from four perspectives.

- **Video-text alignment.** We employ CLIP-T cosine similarity [20] to measure the alignment between generated

Table 1. Summarization of video relighting methods.

Method	Training-Free	Video Base Model	Support Diverse Resolution	Support Background Replacement
AnyPortal	✓	CogVideoX	×	✓
Light-A-Video (A)	✓	AnimateDiff	✓	✓
Light-A-Video (C)	✓	CogVideoX	×	×
Light-A-Video (W)	✓	Wan2.1	✓	×
Lumen	×	Wan2.1	✓	✓
TC-Light	×	VidToMe	✓	×
FlowPortal (Ours)	✓	Wan2.1	✓	✓

frames and the target relighting prompts.

- **Temporal smoothness.** We use CLIP-I [20], the CLIP-based cosine similarity between consecutive frames, to evaluate temporal consistency in the generated videos.
- **Foreground consistency.** As existing metrics rarely address the consistency of primary subjects in relighting tasks, we propose a new evaluation protocol. Specifically, we first extract the foreground region [32] of each output video, then compute (a) **Structural consistency** using SSIM between Canny edge maps [30], (b) **Detail consistency** using PSNR between LoG responses [16] of albedo predictions [2], (c) **Motion consistency** using SSIM between optical flow maps [23], and (d) **Identity consistency** by calculating facial ID similarity [3] for human-centric clips or CLIP-I similarity for others.
- **User preference.** We invite 23 participants to select the best result among four methods based on four criteria: (a) User-Pmt (relevance to the prompt), (b) User-Tmp (temporal coherence), (c) User-Fg (preservation of foreground details and motion), and (d) User-Lit (quality and harmonization of relighting on the foreground).

4.2. Comparison to State-of-the-Art Methods

Qualitative results. Figure 5 shows the qualitative comparison between FlowPortal and other methods. The training-based method Lumen fails to accurately preserve detail fidelity and also exhibits poor lighting alteration and richness. The training-free methods AnyPortal and Light-A-Video produce poor foreground consistency, background quality, and lighting harmonization.

Quantitative results. We construct a test set consisting of 69 pairs of real-world video clips and corresponding relighting prompts with broad diversity, including 54 human-centric videos, 8 animal scenes, and 7 other objects for objective evaluation. Table 2 shows the results among different methods. Our method achieves the best video-text alignment, temporal smoothness, motion consistency, and detail consistency. The training-based method Lumen introduces less variation in lighting conditions with foreground nearly unaltered, which achieves the best detail consistency and identity consistency. Our method ranks the second with negligible consistency value drops, but achieves significant



Figure 5. **Qualitative comparison.** The training-based Lumen exhibits insufficient lighting richness and diversity, with foreground nearly unaltered. The training-free AnyPortal and Light-A-Video show poor structural fidelity and lighting quality. Our method not only maintains structural and detail consistency but also demonstrates high-quality background generation and rich relighting effects.

Table 2. Quantitative and user study results.

Method	CLIP-T	CLIP-I	Structural Consistency	Motion Consistency	Detail Consistency	Identity Consistency	User-Pmt	User-Tmp	User-Fg	User-Lit
AnyPortal	0.3196	0.9817	0.8530	0.8876	40.4853	0.4310	15.3	9.9	8.9	12.1
Light-A-Video (A)	0.2956	0.9684	0.8580	0.8869	40.8727	0.5076	4.7	3.7	3.4	9.4
Lumen	0.3055	0.9746	0.8809	0.8914	40.4193	0.7392	<u>22.7</u>	<u>28.1</u>	<u>35.5</u>	<u>24.4</u>
FlowPortal (Ours)	0.3271	0.9828	<u>0.8804</u>	0.8944	41.2044	<u>0.7328</u>	57.4	58.4	52.2	54.2

lighting adjustment, indicating our method’s high Condition Consistency. For subjective evaluation, Table 2 shows the user preference scores averaged over 17 randomly selected results and 24 participants. Our method achieves the best overall user preference.

Running time. We report the running time of training-free methods on an 80 GB NVIDIA A100 GPU. AnyPortal and Light-A-Video require 20–30 minutes per video due to complex pipelines. In contrast, our method only takes 3–5 minutes across different resolutions roughly equivalent to the direct inference time of a single video diffusion model.

4.3. Ablation Study

We conduct ablation studies to evaluate the effectiveness of Residual-Corrected Flow, High-Frequency Transfer, and the masking mechanism. The qualitative results are shown in Fig. 6, and the quantitative results are presented in Tab. 3.

- **Masking mechanism.** Inference without masking the Consistency Residual Velocity and High-Frequency Transfer causes interference with the generation of the

Table 3. Quantitative ablation study.

Metric	w/o Mask	w/o Residual	w/o High-Frequency	Full
CLIP-T	0.2809	0.3310	<u>0.3290</u>	0.3271
CLIP-I	0.9792	<u>0.9825</u>	0.9798	0.9828
Structural Cons.	0.8649	0.8516	<u>0.8688</u>	0.8804
Motion Cons.	0.8923	<u>0.8933</u>	0.8882	0.8944
Detail Cons.	40.4353	38.5027	<u>40.4852</u>	41.2044
Identity Cons.	<u>0.7123</u>	0.4153	0.5527	0.7328

new background, resulting in an unchanged background structure and poor prompt relevance.

- **Residual-Corrected Flow.** Performing direct inference instead of using the Residual-Corrected Flow leads to severe structural incoherence.
- **High-Frequency Transfer.** The texture-level details fail to be preserved without High-Frequency Transfer due to inaccuracy in structural information and the model’s imperfect controllability under these conditions.

Decoupled Condition Design. We study the effect of our Decoupled Condition Design in Fig. 8. It can be seen that

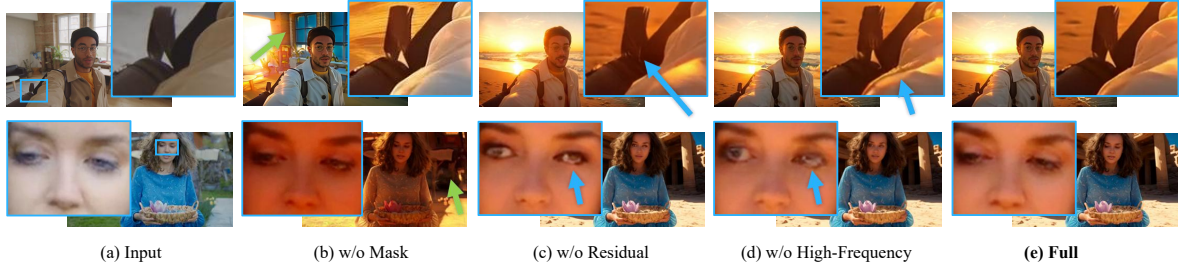


Figure 6. **Ablation Study.** Without mask, background cannot be properly generated in (b). Enlarged local blue regions illustrate the structural inconsistency if (c) removing Residual-Corrected Flow and detail inconsistency if (d) removing High-Frequency Transfer.

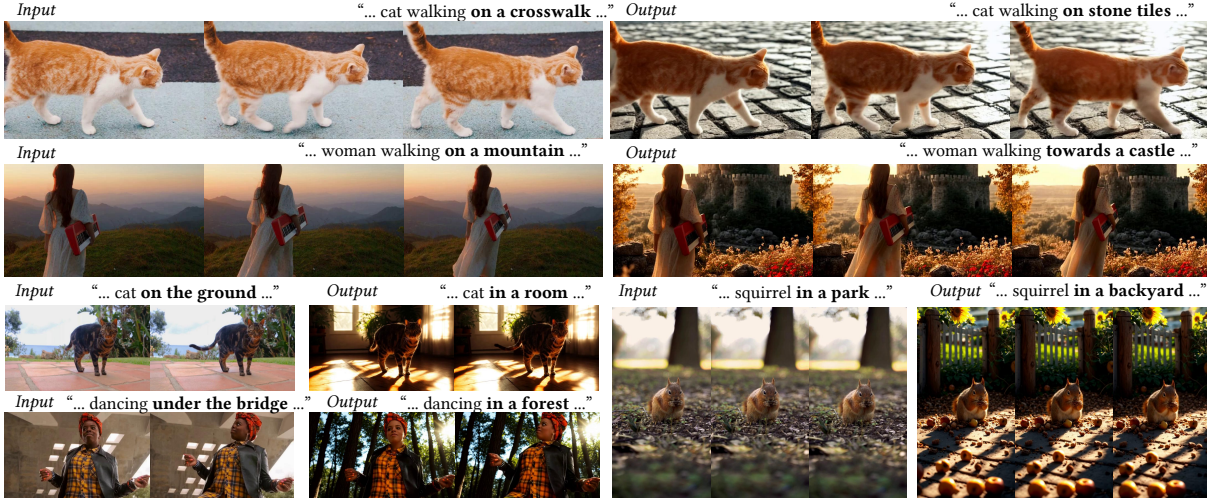


Figure 7. **More visual results.** Our method can generate realistic backlighting effects that simulate light penetrating through clothing, as well as clear shadows cast by objects onto the background under strong lighting conditions.

when using only text conditions, the foreground character fails to be reconstructed properly. Adding structural conditions maintains consistent character structure but fails to produce natural golden lighting. Incorporating the reference frame results in stronger and prompt-consistent lighting, but leads to incomplete character structure. Only when all conditions are utilized, does our method achieve the most natural results in both structure and lighting.

5. Conclusion and Discussion

In this paper, we propose FlowPortal, a novel training-free framework for efficient video relighting and background replacement. We introduce a novel Residual-Corrected Flow mechanism with Decoupled Condition Design that enforces Condition Consistency. A High-Frequency Transfer module is designed to enhance the detail fidelity and a masking mechanism is applied to isolate background regions for high-quality generation. Both qualitative and quantitative experiments demonstrate that our method achieves superior background generation quality, foreground consistency, and lighting realism.

Although our method performs well in most cases, it



Figure 8. **Ablation on Decoupled Condition Design.** The absence of structural information or the reference frame leads to degraded structural fidelity and unnatural illumination.

may still be constrained by the generative capability of the underlying base models (*i.e.*, IC-Light for relighting the reference frame and Wan2.1 for background generation).

It is evident that our Residual-Corrected Flow framework is not exclusively designed for relighting. A potential future direction could be the exploration of extending it to various video editing tasks, including video stylization, colorization, object manipulation, human editing, and even more general editing tasks. Further algorithmic improvements on the Residual-Corrected Flow, reconstruction rather than relying on direct directional addition, would be another promising future direction.

References

- [1] AIGC-Apps. Videox-fun: Github repository, 2025. Github repository: github.com/aigc-apps/VideoX-Fun. 4, 6
- [2] Chris Careaga and Yağız Aksoy. Colorful diffuse intrinsic image decomposition in the wild. *ACM TOG*, 43(6):1–12, 2024. 6
- [3] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *CVPR*, pages 4690–4699, 2019. 6
- [4] Ye Fang, Zeyi Sun, Shangzhan Zhang, Tong Wu, Yinghao Xu, Pan Zhang, Jiaqi Wang, Gordon Wetzstein, and Dahua Lin. Relightvid: Temporal-consistent diffusion model for video relighting. *arXiv preprint arXiv:2501.16330*, 2025. 2, 6
- [5] Wenshuo Gao, Xicheng Lan, and Shuai Yang. Anyportal: Zero-shot consistent video background replacement. In *ICCV*, pages 18990–18999, 2025. 2, 6
- [6] Kaiwen Guo, Peter Lincoln, Philip Davidson, Jay Busch, Xueming Yu, Matt Whalen, Geoff Harvey, Sergio Orts-Escolano, Rohit Pandey, Jason Dourgarian, et al. The relightables: Volumetric performance capture of humans with realistic relighting. *ACM TOG*, 38(6):1–19, 2019. 2
- [7] Kai He, Ruofan Liang, Jacob Munkberg, Jon Hasselgren, Nandita Vijaykumar, Alexander Keller, Sanja Fidler, Igor Gilitschenski, Zan Gojcic, and Zian Wang. Unirelight: Learning joint decomposition and synthesis for video relighting. *arXiv preprint arXiv:2506.15673*, 2025. 2
- [8] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross-attention control. In *ICLR*, pages 1–12, 2022. 5
- [9] Hoon Kim, Minje Jang, Wonjun Yoon, Jisoo Lee, Donghyun Na, and Sanghyun Woo. Switchlight: Co-design of physics-driven architecture and pre-training framework for human portrait relighting. In *CVPR*, pages 25096–25106, 2024. 2
- [10] Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024. 3
- [11] Vladimir Kulikov, Matan Kleiner, Inbar Huberman-Spiegelglas, and Tomer Michaeli. Flowedit: Inversion-free text-based editing using pre-trained flow models. In *ICCV*, pages 19721–19730, 2025. 5
- [12] Yuanze Lin, Yi-Wen Chen, Yi-Hsuan Tsai, Ronald Clark, and Ming-Hsuan Yang. Illumicraft: Unified geometry and illumination diffusion for controllable video generation. *arXiv preprint arXiv:2506.03150*, 2025. 2
- [13] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022. 3
- [14] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. *arXiv preprint arXiv:2209.03003*, 2022. 3
- [15] Yang Liu, Chuanchen Luo, Zimo Tang, Yingyan Li, Yuran Yang, Yuanyong Ning, Lue Fan, Junran Peng, and Zhaoxiang Zhang. Tc-light: Temporally consistent relighting for dynamic long videos. *arXiv preprint arXiv:2506.18904*, 2025. 2, 6
- [16] David Marr and Ellen Hildreth. Theory of edge detection. *Proceedings of the Royal Society of London. Series B. Biological Sciences*, 207(1167):187–217, 1980. 6
- [17] Yiqun Mei, Mingming He, Li Ma, Julien Philip, Wenqi Xian, David M George, Xueming Yu, Gabriel Dedic, Ahmet Levent Taşel, Ning Yu, et al. Lux post facto: Learning portrait performance relighting with conditional video diffusion and a hybrid dataset. In *CVPR*, pages 5510–5522, 2025. 2, 6
- [18] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *CVPR*, pages 6038–6047, 2023. 5
- [19] Rohit Pandey, Sergio Orts-Escolano, Chloe Legendre, Christian Haene, Sofien Bouaziz, Christoph Rhemann, Paul E Debevec, and Sean Ryan Fanello. Total relighting: learning to relight portraits for background replacement. *ACM TOG*, 40(4):43–1, 2021. 2
- [20] Chenyang Qi, Xiaodong Cun, Yong Zhang, Chenyang Lei, Xintao Wang, Ying Shan, and Qifeng Chen. Fatezero: Fusing attentions for zero-shot text-based video editing. In *ICCV*, pages 15932–15942, 2023. 6
- [21] Mengwei Ren, Wei Xiong, Jae Shin Yoon, Zhixin Shu, Jianming Zhang, HyunJoon Jung, Guido Gerig, and He Zhang. Relightful harmonization: Lighting-aware portrait background replacement. In *CVPR*, pages 6452–6462, 2024. 2
- [22] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. 5
- [23] Zachary Teed and Jia Deng. Raft: Recurrent all-pairs field transforms for optical flow. In *ECCV*, pages 402–419. Springer, 2020. 6
- [24] Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. 3, 6
- [25] Chen Henry Wu and Fernando De la Torre. A latent space of stochastic diffusion models for zero-shot image editing and guidance. In *ICCV*, pages 7378–7387, 2023. 5
- [26] Xiaoyan Xing, Konrad Groh, Sezer Karaoglu, Theo Gevers, and Anand Bhattad. Luminet: Latent intrinsics meets diffusion models for indoor scene relighting. In *CVPR*, pages 442–452, 2025. 2
- [27] Peiqing Yang, Shangchen Zhou, Jixin Zhao, Qingyi Tao, and Chen Change Loy. Matanyone: Stable video matting with consistent memory propagation. In *CVPR*, pages 7299–7308, 2025. 6
- [28] Jianshu Zeng, Yuxuan Liu, Yutong Feng, Chenxuan Miao, Zixiang Gao, Jiwang Qu, Jianzhang Zhang, Bin Wang, and Kun Yuan. Lumen: Consistent video relighting and harmonious background replacement with video generative models. *arXiv preprint arXiv:2508.12945*, 2025. 2, 6

- 642 [29] Longwen Zhang, Qixuan Zhang, Minye Wu, Jingyi Yu, and
643 Lan Xu. Neural video portrait relighting in real-time via con-
644 sistency modeling. In *ICCV*, pages 802–812, 2021. [2](#)
- 645 [30] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding
646 conditional control to text-to-image diffusion models. In
647 *ICCV*, pages 3836–3847, 2023. [4](#), [6](#)
- 648 [31] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Scaling
649 in-the-wild training for diffusion-based illumination harmo-
650 nization and editing by imposing consistent light transport.
651 In *ICLR*, pages 1–12, 2025. [2](#), [4](#), [6](#)
- 652 [32] Peng Zheng, Dehong Gao, Deng-Ping Fan, Li Liu, Jorma
653 Laaksonen, Wanli Ouyang, and Nicu Sebe. Bilateral referen-
654 ce for high-resolution dichotomous image segmentation.
655 *CAAI Artificial Intelligence Research*, 2024. [6](#)
- 656 [33] Peng Zheng, Dehong Gao, Deng-Ping Fan, Li Liu, Jorma
657 Laaksonen, Wanli Ouyang, and Nicu Sebe. Bilateral referen-
658 ce for high-resolution dichotomous image segmentation.
659 *arXiv preprint arXiv:2401.03407*, 2024. [6](#)
- 660 [34] Yujie Zhou, Jiazi Bu, Pengyang Ling, Pan Zhang, Tong
661 Wu, Qidong Huang, Jinsong Li, Xiaoyi Dong, Yuhang
662 Zang, Yuhang Cao, et al. Light-a-video: Training-free
663 video relighting via progressive light fusion. *arXiv preprint*
664 *arXiv:2502.08590*, 2025. [2](#), [6](#)